Question:
Assume that B Corporation has the following total profit function equation: Total Profit = \pi = - $3,000 - $2,400 Q + $350Q^2 - $8.333Q^3 a) Compute the marginal profit b) Determine its output quantities
Answer Choices:
A. 22 and 7 units
B. 21 and 8 units
C. 23 and 6 units
D. 26 and 2 units
E. 24 and 4 units
F. 25 and 3 units
G. 20 and 5 units
H. 27 and 1 unit
I. 18 and 10 units
J. 19 and 9 units

Reasoning + Answer:

<reasoning process here>
We are given the total profit function:

ÏĢ = -3000 - 2400Q + 350QÂ² - 8.333QÂ³

a) To compute the marginal profit, we take the derivative of the total profit function with respect to Q:

Marginal Profit (MP) = dÏĢ/dQ

So,

MP = d/dQ [-3000 - 2400Q + 350QÂ² - 8.333QÂ³]

Differentiate term by term:

- d/dQ (-3000) = 0
- d/dQ (-2400Q) = -2400
- d/dQ (350QÂ²) = 700Q
- d/dQ (-8.333QÂ³) = -25QÂ² (since 8.333 ÃĹ 3 = 25)

So,

Marginal Profit = -2400 + 700Q - 25QÂ²

That âĢĻs the marginal profit function.

b) To determine the output quantities, we need to find the quantity Q where marginal profit is zero (since that's where profit is maximized, assuming the firm is a profit-maximizing firm).

Set MP = 0:

-2400 + 700Q - 25QÂ² = 0

Multiply both sides by -1 to make it easier:

2400 - 700Q + 25QÂ² = 0

Rewrite:

25QÂ² - 700Q + 2400 = 0

We can simplify this equation by dividing all terms by 25:

QÂ² - 28Q + 96 = 0

Now solve the quadratic equation:

Q = [28 Â± âĪļ(28Â² - 4ÃĹ1ÃĹ96)] / 2
= [28 Â± âĪļ(784 - 384)] / 2
= [28 Â± âĪļ400] / 2
= [28 Â± 20] / 2

So:

Q = (28 + 20)/2 = 48/2 = 24
Q = (28 - 20)/2 = 8/2 = 4

Wait âĢĶ this gives us two values: Q = 24 and Q = 4.

But the answer options are pairs like "22 and 7 units", "21 and 8 units", etc.

This suggests that the output quantities are two values, perhaps both are valid solutions? But typically, for profit maximization, only one Q is optimal.

Wait âĢĶ but we found that marginal profit is zero at Q = 24 and Q = 4?

But check: the profit function is cubic, and if the marginal profit is zero at two points, then these are turning points. One may be a maximum and one a minimum.

But since the profit function has negative cubic term (-8.333QÂ³), profit will eventually decrease. So likely, one of these is a local maximum.

We should verify which of these gives a maximum (i.e., check second derivative):

Second derivative (dÂ²ÏĢ/dQÂ²) = derivative of marginal profit:

MP = -2400 + 700Q - 25QÂ²
dÂ²ÏĢ/dQÂ² = 700 - 50Q

At Q = 4:

dÂ²ÏĢ/dQÂ² = 700 - 200 = 500 > 0 âĨĴ local minimum

At Q = 24:

dÂ²ÏĢ/dQÂ² = 700 - 50ÃĹ24 = 700 - 1200 = -500